

Madhav Makwana

Software Engineer | LLM Applications · RAG Systems · Agentic AI | DTU 2026

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Education

Delhi Technological University

2022 – 2026

B.Tech in Engineering Physics (Electronics), Minor in Computer Science

Experience

Appsoleut Games

June 2025 – Aug 2025

Data Science Intern

Gurugram

- Analyzed **10K+ gameplay sessions** to map early-player behavior patterns (retries, collisions, session duration) and identify high-churn business segments for product intervention.
- Trained a logistic regression classifier on engineered gameplay features, achieving **AUC > 0.80** on hold-out data.
- Delivered a **3-tier risk segmentation model** (low/medium/high) to guide onboarding strategy and retention decisions.

Projects

ResearchMind — Multi-Agent AI Research System

[🔗 Live Demo](#) | [🐙 GitHub](#)

Python, FastAPI, LangChain, LangGraph, MCP, OpenAI API, Tavily API, Docker, Prometheus

- Architected a multi-agent research pipeline on **LangGraph** — planner, parallel researchers fanned out via the **Send API** and routed through a custom **MCP tool server**, writer, and critic loop — for grounded report generation.
- Engineered a standalone **MCP tool server** (web search, scraping, arXiv, calculator) shared by tool-calling researcher and verifier agents, each selecting tools independently.
- Designed a self-correcting critic loop paired with an independent **citation verifier** that re-checks a report's load-bearing claims against fresh sources, reducing hallucination risk.
- Designed a **planner** that scales sub-question count with topic complexity or asks for clarification when a topic is too ambiguous to research directly.

YTLens — RAG System with Evaluated Retrieval Pipeline

[🔗 Live Demo](#) | [🐙 GitHub](#)

Python, FastAPI, LangChain, Pinecone, Cross-Encoder Reranking, JWT, SQLAlchemy

- Constructed a 25-question **evaluation harness** comparing dense, hybrid BM25+dense (RRF), and **cross-encoder reranking** — reranking cut top-3 retrieval misses by 33%, driving a leaner dense+rerank production design.
- Migrated vector storage from FAISS to **Pinecone** — eliminates memory growth, persists videos across redeploy.
- Implemented a **3-layer transcript fetching strategy** (direct, then proxy, then manual fallback) ensuring zero hard failures.
- Added **SSE streaming** for token-by-token answers, with bcrypt-based JWT auth and Dockerized deployment on Render.

LLM Gateway — Secure, Cost-Aware Routing Proxy for LLM APIs

[🐙 GitHub](#)

Python, FastAPI, Redis, FAISS, Docker, AWS EC2, Prometheus, Grafana, pytest

- Built an OpenAI-compatible gateway with circuit-breaker routing across Groq and OpenAI, auto-failing over on timeouts.
- Implemented prompt injection/jailbreak detection via embedding similarity, blocking adversarial prompts pre-inference.
- Added dual-layer caching (Redis exact-match + FAISS semantic) to cut redundant LLM calls and inference cost.
- Enforced per-key budget caps via a cost ledger; deployed on AWS EC2 with Prometheus/Grafana observability.

Skills

Languages: Python, C++, SQL

CS Fundamentals: Data Structures & Algorithms, OOP, DBMS, Operating Systems

Backend: FastAPI, REST APIs, AsyncIO, Pydantic, SQLAlchemy, JWT Auth, OAuth2, Middleware, Logging

Generative AI & LLM Systems: LangChain, LangGraph, MCP, RAG, Semantic Caching, OpenAI, Groq, Agentic AI

Databases & Caching: MySQL, Redis, SQLite, FAISS, ChromaDB

DevOps & Infra: Docker, AWS EC2, Prometheus, Grafana, Git, GitHub, Render, Locust

Machine Learning: scikit-learn, TensorFlow, Keras, HuggingFace

Tools: Streamlit, Postman, Linux

Achievements

- Ranked among **top 5%** in NPTEL Introduction to Database (Q4 toppers list) [\[CERTIFICATE\]](#) [🔗](#)
- Awarded **Proficiency in Computer Science** in Class 12 [\[CERTIFICATE\]](#) [🔗](#)
- SOF Olympiads (top ~5% globally, qualified Level 2): **IEO Delhi Rank 19** [\[CERT\]](#) [🔗](#); **IMO Int. Rank 4033** [\[CERT\]](#) [🔗](#); **NSO Int. Rank 5065** [\[CERT\]](#) [🔗](#)